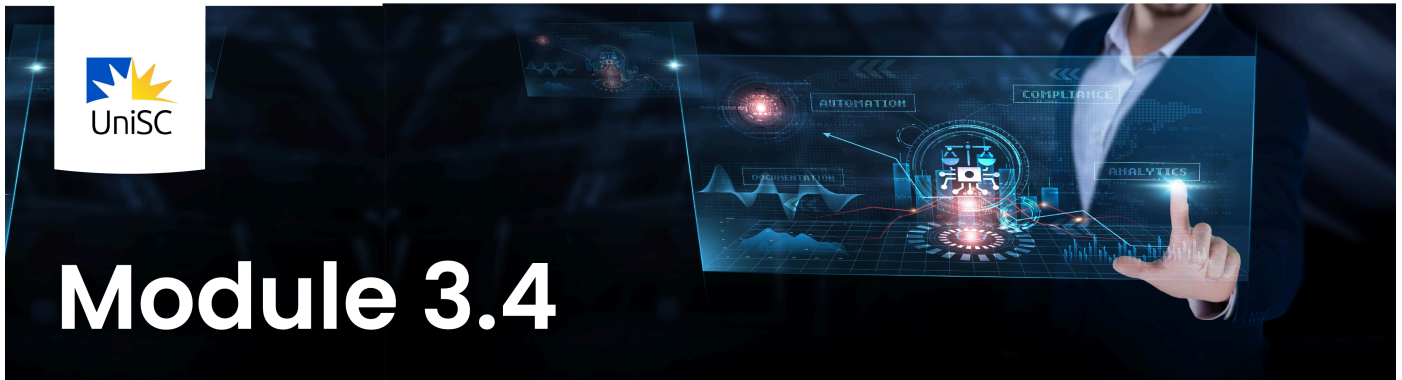


3.4 Real Returns and Purchasing Power

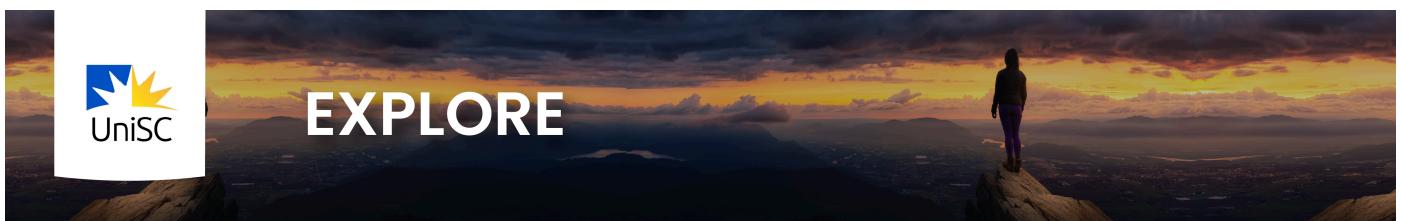


Real Returns and Inflation



Guiding Questions

- Why can nominal returns be misleading when evaluating investment performance?
- How does inflation affect the real value of investment returns?
- How can we convert nominal returns into real returns?
- Why are real returns particularly important for long-term investors?



What will you learn?

You will learn:

- The difference between nominal and real returns.
- How inflation reduces the purchasing power of investment gains.
- How to calculate real returns using the Fisher equation.

- Why evaluating investments in real terms is critical for long-term wealth planning.
-

1. Why Does Inflation Matter for Investors?

In previous sections we examined how to measure portfolio returns, risk, and risk-adjusted performance. However, another important factor must be considered when evaluating investment outcomes: inflation.

Inflation refers to the general increase in prices of goods and services in an economy over time. When inflation rises, the purchasing power of money declines, meaning that a given amount of money can buy fewer goods and services than before.

For investors, this means that the true value of investment returns depends not only on how much money is earned, but also on how prices change during the same period.

For example, if an investment generates a return of 10% while inflation is 8%, the real improvement in purchasing power is only about 2%.

Understanding inflation is therefore important because:

- inflation reduces the purchasing power of investment returns
- nominal returns may overstate actual economic gains
- long-term investment decisions must consider real wealth growth
- comparing returns across time requires adjusting for inflation

Because of these reasons, investors often distinguish between **nominal returns** and **real returns**.



Watch

[Real and Nominal Returns](#)

Now add inflation: apples jump to \$6

A poor harvest and visa restrictions on seasonal pickers shrink the apple supply - prices rise 20%. You still get your \$110 back.

TODAY	ONE YEAR LATER - apples now \$6
\$100 → 20 apples 	\$110 ÷ \$6 = 18.33 apples  (18.33 - that's 1.67 apples short of where you started)

18.33 apples is less than the 20 you started with. Your real return is negative, even though your nominal return was a positive 10%.

Details
Transcript

⋮

0:00 Now let's talk about a simple but easily missed

0:02 idea the difference between a real re- turn and a

0:06 nominal return. Inflation erodes the value of

0:10 money, and the value of money is not the number

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2. Nominal Returns vs Real Returns

A **nominal return** is the percentage change in the value of an investment measured in money terms. It does not account for inflation.

A **real return**, in contrast, measures the increase in purchasing power after adjusting for inflation.

The relationship between these two measures can be expressed conceptually as:

Real return = Nominal return – Inflation (approximately)

However, for more accurate calculations, a precise formula is used, which accounts for the compounding effects of inflation.

3. The Fisher Equation

The relationship between nominal return, real return, and inflation is described by the **Fisher equation**.

Fisher Equation

$$r_{\text{real}} = \frac{1+r_{\text{nominal}}}{1+\text{inflation}} - 1$$

Where:

r_{nominal} = nominal investment return

r_{real} = real return

inflation = inflation rate

Rearranging the formula allows us to calculate the real return:

$$r_{\text{real}} = \frac{1+r_{\text{nominal}}}{1+\text{inflation}} - 1$$

This equation shows that the real return is slightly lower than the simple difference between nominal return and inflation because both variables are compounded.

4. Example: Calculating Real Return

Suppose a portfolio generated a **nominal return of 12%**, while the inflation rate during the same period was **5%**.

Step 1: Apply the Fisher equation

$$r_{\text{real}} = \frac{1.12}{1.05} - 1$$

Step 2: Calculate the result

$$r_{\text{real}} = 1.0667 - 1 = 6.67\%$$

Interpretation

Although the portfolio produced a 12% nominal return, the actual improvement in purchasing power was only 6.67%.

This example illustrates that inflation significantly reduces the real value of investment gains.

5. Inflation and Long-Term Investment Performance

Inflation becomes especially important for long-term investments.

Even moderate inflation rates can significantly reduce purchasing power over time.

Consider the following example.

Suppose an investment grows at 8% per year, while inflation averages 4% per year.

In nominal terms, the investment appears to grow rapidly. However, when adjusted for inflation, the real growth rate is closer to 4% per year.

Over long periods, this difference becomes substantial.

For instance:

\$10,000 invested for 20 years at 8% grows to approximately \$46,610.

However, if inflation averages 4%, the real purchasing power of that amount is closer to \$21,900 in today's dollars.

This demonstrates why investors should evaluate long-term investment performance in real terms rather than nominal terms.

6. Inflation and Investment Decision-Making

Different types of investments respond differently to inflation.

Some assets tend to perform relatively well during inflationary periods, while others may lose value.

Examples include:

Assets that may provide inflation protection

- real estate
- commodities
- inflation-indexed bonds
- certain equities

Assets that may be negatively affected by inflation

- fixed-rate bonds
- cash savings
- long-term fixed-income securities

Investors therefore often consider inflation expectations when constructing portfolios.

Portfolio managers may include assets that **hedge against inflation risk** in order to protect the real value of wealth.

7. Interpreting Real Returns in Portfolio Analysis

When evaluating investment performance, real returns provide a more accurate picture of **wealth creation**.

Two portfolios may have identical nominal returns, but if one operates in a period of higher inflation, its real performance will be weaker.

For this reason:

- long-term investors focus primarily on real returns
- pension funds and retirement portfolios emphasize purchasing power
- financial planning often incorporates expected inflation rates

Real return analysis therefore complements the earlier measures of portfolio return, risk, and risk-adjusted performance.

8. Key Insights from Real Return Analysis

From this section we learn:

- Inflation reduces the purchasing power of money over time
- Nominal returns measure gains in monetary terms only
- Real returns measure changes in purchasing power
- The Fisher equation links nominal returns, real returns, and inflation
- Long-term investment evaluation should focus on real returns rather than nominal returns
- Inflation expectations play an important role in portfolio construction

Understanding real returns helps investors assess whether their investments are truly increasing wealth in economic terms.

Reflect

What is the difference between nominal returns and real returns?

Why can nominal returns sometimes be misleading for investors?

What does inflation represent in economic terms?

Why is it important for long-term investors to consider real returns?

How does inflation influence portfolio allocation decisions?

Solve

Question 1

An investment generates a nominal return of 10%, while inflation is 4%.

Calculate the real return using the Fisher equation.

Question 2

A portfolio produces a nominal return of 15% during a year when inflation is 6%.

Calculate the real return.

Question 3

Suppose an investment generates a nominal return of 7% but inflation rises to 8%.

What is the real return?

Explain the result.

Question 4

Explain why inflation can have a significant impact on long-term investment performance.

Answers available [here \(https://learn.usc.edu.au/courses/35359/files/3292621?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3292621?wrap=1).

Critically evaluate

Two investors each earn a nominal return of 9% on their portfolios.

Investor A lives in a country where inflation is 2%, while Investor B lives in a country where inflation is 7%.

Which investor experiences the higher real return? Explain the implications for wealth accumulation.